

Visualization and the Science/Art of Making the Complex Simple

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ABSTRACT

Visualization, whether in 2D or 3D, is frequently faced with the challenge of navigating complexity, be it in the realm of data, design, or broader concept. This panel aims to delve into the intricacies of effectively communicating ideas through visualization, drawing upon the diverse experiences and expertise of panelists representing esteemed institutions such as NASA, NCSA, and The New York Times. By exploring the challenges encountered and the solutions devised by these industry leaders, the panel aims to offer valuable insights into the art and science of conveying complex ideas through visualization.

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1 INTRODUCTION

Though graphical animation has been around for a long time, visualizing complex concepts or data presents its own challenges. Representing something that is often abstract in a scientifically accurate manner takes a lot of iterations to get right. The stress on accuracy and precision is something that is unique and common to both science and journalism. This panel will start with some of the work of the panelists from NASA, NCSA, and The New York Times to set the tone. It will address some of the techniques to deal with complexity and will end with some time for Q&A.

2 PUBLISHED WORK

Some of the work we'll discuss:

2.1 Masks Work. Really. We'll Show You How

This project allowed readers to see clearly the protection masks can provide, turning an abstract notion of safety into a tangible visual explanation. The reporting was based on interviews with leading aerosol experts, and the backbone of the presentation was a 3-D animation at the particle scale, which let readers see the

individual fibers of a mask and an N95 respirator. The animation moved through the physics of filtration at work at a microscopic level to show that even though the material of these masks looks thin, they are able to trap viral particles effectively. [Fleisher et al., 2020].

2.2 Why Opening Windows Is a Key to Reopening Schools

As schools considered reopening for in-person learning, we built on our experience [Bartzokas et al., 2021] simulating particles and airflow to illustrate the simple steps schools could take to reduce coronavirus exposure. We gained access to a Harlem classroom to take precise measurements and construct a 3-D model of the space. We worked with an engineering firm to show how air would move around the room under different scenarios, how infectious aerosols might build up in the space, and, crucially, how to reduce them. Reporting with leading experts in ventilation and New York City's school reopening plan helped place the data in context.

2.3 See How the Dixie Fire Created Its Own Weather

Days after California's Dixie fire [Popovich et al., 2021] ignited in July 2021, towering storm clouds burst from the flames, generating lightning and wild winds. By merging data from radar stations in California and Nevada, we were able to reconstruct Dixie's first thunderclouds in 3-D. Terrain, fire, lightning and other data sets were then designed, layered together and rendered out, revealing how the wildfire fed its own destructive spread.

2.4 A Journey to the Center of the Milky Way: Stellar Orbits around its Central Black Hole

This visual tour summarizes results from Nobel laureate Andrea Ghez's study of the motions of stars around the Milky Way's central black hole, revealing a rich and surprising environment. Created by a "Renaissance Team" of artists, programmers, astrophysicists, and social scientists, this cinematic scientific visualization required fusing multiple datasets across vast scale changes, custom software development, camera choreography, GPU-accelerated rendering on the Delta supercomputer, and focus groups & surveys with test audiences. [Advanced Visualization Lab, 2023]

3 CONCLUSION

In an era characterized by uncertain times and overwhelming information, we managed to distill a critical public health issue into an engaging and visually compelling narrative. Our unique visual

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strategy forged a connection between millions of readers and emerging research, transforming the complex into the comprehensible through intuitive, three-dimensional interaction.

Our journey to deliver this vital public health information explored various methodologies, leading to groundbreaking advancements in visual storytelling. We engineered a bespoke system that seamlessly translated our spatial reporting and modeling across multiple platforms: an immersive 3-D experience on the web, an intricate 2-D print graphic, and an augmented reality effect tailored for our Instagram audience.

The project's technical challenges were monumental, yet the end result was a remarkably smooth and accessible experience for readers. By achieving this delicate balance, we not only advanced the art and science of visualization but also laid a robust foundation for future storytelling endeavors at The Times.

Notes: The Dixie fire's smoke and clouds were created using reflectivity data from three Next-Generation Radar network stations collected on July 19, 2021, between 11:00 a.m. and 11:00 p.m, Pacific Daylight Time.

The raw data was collected every 10 minutes in radial sweeps around the radar stations, each at a higher altitude. The Times combined and reformatted the data using Py-ART, a collection of algorithms and utilities used regularly in radar analysis. We then filtered it to reduce noise.

We applied color and texture to the 3-D volume to approximate a smoke- and cloud-like look. And we interpolated the sequence in time to create a smoother video animation.

The radar volume rendering was based on previous work by Dr. Lareau of the University of Nevada, Reno.

The lightning strikes were recorded by the Vaisala Lightning Detection System. We only showed strikes that reached the ground.

The terrain was created using topographic data from the U.S.G.S. 3-D Elevation Program and color imagery from the Copernicus Sentinel-2 satellites, retrieved at a resolution of 10 meters. To eliminate 2-D cloud cover on the landscape, we generated a composite of satellite images from June 1 to June 30, 2021.

The fire footprint was created by combining perimeter data from the California Department of Forestry and Fire Protection along with fire detections data from NASA's Fire Information for Resource Management System and the U.S. Forest Service Active Fire Mapping Program.

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